

DINESH KUMAR M

Software Engineer | Backend Engineer

Java · Spring Boot · Rust · Backend Systems · Microservices · Distributed Systems

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PROFESSIONAL SUMMARY

Backend-focused Software Engineer with 2 years of experience designing, building, deploying, and supporting production backend systems for a gaming platform serving 20K+ registered users. Experienced in developing Java monolithic applications and Rust-based microservices, delivering scalable backend solutions through API design, database modeling, distributed service communication, caching strategies, and production support.

Hands-on experience designing REST APIs, GraphQL services, and gRPC-based communication while working across the complete Software Development Life Cycle (SDLC), from requirement analysis and solution design to implementation, deployment validation, production monitoring, and incident resolution. Strong foundation in PostgreSQL, Redis, DynamoDB, distributed systems, backend performance optimization, and cross-functional collaboration within Agile product teams.

TECHNICAL SKILLS

Languages: Java · Rust · TypeScript · JavaScript · SQL

Backend Technologies: Spring Framework · Spring Boot · Spring MVC · Spring Data JPA · NestJS · Node.js

Frontend: Next.js · React · Angular

APIs & Communication: REST APIs · GraphQL · Apollo GraphQL · gRPC · Protocol Buffers

Databases: PostgreSQL · Redis · DynamoDB · MySQL · MongoDB · NeonDB

Backend Engineering: Backend Development · API Design · Database Design · Data Modeling · SQL Query Optimization · Authentication · Authorization · RBAC · Redis Caching · Performance Optimization · Production Support · Root Cause Analysis · Distributed Systems · Microservices · Monolithic Architecture · Event-Driven Architecture · System Design

DevOps & Developer Tools: Git · Docker · Jenkins · ArgoCD · Grafana · Maven · Postman · Altair GraphQL · RedisInsight · pgAdmin · NoSQL Workbench · Termius · Warp

PROFESSIONAL EXPERIENCE

Aretedge Innovations Pvt. Ltd.

Associate Software Engineer

Hyderabad, Telangana, India (On-site)

Aug 2024 – May 2026

Domain: Gaming Platform | **Architecture:** Java Monolith → Rust Microservices

Tech Stack: Java · Spring Boot · Rust · NestJS · PostgreSQL · Redis · DynamoDB · GraphQL · gRPC · Docker · Jenkins · ArgoCD · Grafana · Next.js

Feature Ownership

- **Owned the end-to-end backend implementation of the Challenge Module**, translating business requirements into production-ready backend workflows by collaborating with architects on solution design, designing APIs, implementing business logic, modeling databases, validating deployments, and providing post-release production support.
- **Engineered backend services for the Casual Games Module**, integrating external gaming provider APIs and implementing gameplay, participation, rewards, and leaderboard workflows that aligned technical implementation with business requirements.
- **Designed and implemented Double Elimination tournament workflows**, translating complex tournament rules into maintainable backend logic that became the foundation for tournament functionality.

Backend Engineering

- **Designed and delivered 20+ production backend APIs** across Java monolithic applications and Rust microservices using REST APIs, GraphQL, and gRPC, supporting Challenge, Casual Games, Tournament, Gateway, Membership, Authentication, Notification, Metadata, and Event services.
- Designed API contracts, implemented business workflows, integrated service-to-service communication, and collaborated with architects to review technical approaches before implementation.
- Developed GraphQL schemas, resolvers, gRPC services, and Protocol Buffer definitions to enable efficient communication between distributed backend services.
- Implemented JWT-based authentication, authorization, and Role-Based Access Control (RBAC) to secure player and administrative operations across backend services.

- Integrated third-party gaming provider APIs into platform workflows while ensuring reliable communication between internal and external services.

Database Engineering

- Designed PostgreSQL schemas, relational data models, indexes, JSONB structures, materialized views, and database functions supporting transactional gaming workflows while optimizing SQL queries for maintainability and efficient data access.
- Designed DynamoDB access patterns by selecting partition keys, sort keys, and Global Secondary Indexes (GSIs) based on application query requirements while adhering to established architectural conventions.
- Developed Node.js migration scripts to transform and migrate application data from DynamoDB into optimized PostgreSQL schemas during backend modernization efforts.

Performance & Scalability

- Implemented Redis caching strategies using TTL and database fallback mechanisms to reduce repeated database access for leaderboard and other high-read workflows while maintaining data consistency.
- Improved backend maintainability by simplifying SQL query logic, reducing redundant parameters, and leveraging relational database design to minimize unnecessary data duplication.

Production Engineering

- Supported **25+ production releases** by validating backend functionality, coordinating with QA, managing development and pre-production deployments through Jenkins pipelines, and contributing to controlled production rollouts.
- Resolved **10+ production incidents** by reproducing reported issues, analyzing service logs in ArgoCD, investigating PostgreSQL and Redis interactions, identifying root causes, implementing validated fixes, and supporting production verification.
- Delivered **8+ production hotfixes**, coordinating with QA, technical leads, and DevOps teams to ensure stable releases with minimal impact on production systems.

Engineering Collaboration

- Participated in architecture discussions, solution reviews, sprint planning, story estimation, database design reviews, API design discussions, and Agile ceremonies while collaborating with architects, backend engineers, QA, DevOps, and product stakeholders.
- Reviewed merge requests, contributed to code quality improvements, assisted teammates with development and QA environment issues, and supported knowledge sharing through technical discussions.
- Authored API documentation, feature design documents, and technical specifications to improve engineering collaboration and accelerate onboarding.
- Contributed throughout the complete Software Development Life Cycle (SDLC), including requirement analysis, implementation planning, API testing, deployment validation, production monitoring, incident resolution, and post-release support.

INTERNSHIP & TRAINING

Aspire Systems

Graduate Engineer Trainee

Graduate Engineering Intern

Graduate Engineer Trainee

Chennai, Tamil Nadu, India (Remote)

Apr 2023 – Jul 2024

Jan 2023 – Mar 2023

Sep 2022 – Dec 2022

Training Focus: Enterprise Software Engineering

Tech Stack: Java · Spring Boot · Angular · Node.js · MySQL · REST APIs · Git

Enterprise Software Engineering Training

- Completed structured enterprise software engineering training focused on Java, Spring Boot, Angular, Node.js, MySQL, object-oriented programming, software architecture, and the Software Development Life Cycle (SDLC) through classroom learning and project-based development.
- Built enterprise-style full-stack applications by implementing frontend features, backend business logic, REST APIs, relational database operations, debugging, API testing, and issue resolution while following enterprise development practices.
- Applied object-oriented programming, layered architecture, modular application design, relational database design, version control, Agile development practices, and collaborative software engineering principles through project-based assignments.

PROJECTS

Personal Finance Tracker | *Next.js · TypeScript · PostgreSQL · NeonDB · Zustand*

Links: [Live Demo](#) | [GitHub](#)

- Designed and developed a personal finance management application to centralize expense tracking, lending, borrowing, investments, repayment management, and financial analytics through a unified dashboard.
- Implemented secure authentication, responsive user interfaces, Row-Level Security (RLS), and API-driven workflows to ensure secure access and reliable financial data management.
- Improved application responsiveness by replacing an earlier Supabase + Drizzle ORM implementation with a simplified Neon PostgreSQL architecture after evaluating authentication latency and architectural trade-offs.
- Implemented CSV import/export functionality and categorized financial transactions to support reporting and long-term expense analysis.

Student Management System | *Java · Angular · MySQL*

Links: [Live Demo](#) | [GitHub \(Angular\)](#) | [GitHub \(Java\)](#)

- Developed a role-based academic management system supporting attendance, examinations, fee management, and administrative workflows.
- Designed relational database schemas and implemented REST APIs with Java to support modular backend services and efficient data management.
- Implemented Role-Based Access Control (RBAC) and progressively evolved the application from frontend prototypes into a backend-driven system to strengthen full-stack engineering practices.

E-Commerce Platform | *Spring Boot · Spring Data JPA · MySQL*

Links: [Live Demo](#) | [GitHub \(Frontend\)](#) | [GitHub \(Spring Boot\)](#)

- Built a layered Spring Boot backend implementing REST APIs, Spring Data JPA, and relational database modeling to support product, shopping cart, and order management workflows.
- Applied clean architecture principles, service-layer abstraction, and persistent data management to evolve the project from a static prototype into a backend-driven application.
- Strengthened practical understanding of Spring Boot application structure, dependency injection, repository design, and relational data access.

Travel Agency Platform | *Angular · Node.js · MongoDB*

Links: [Live Demo](#) | [GitHub](#)

- Built the architectural foundation for a modular travel management platform by implementing authentication, backend services, and MongoDB-backed data models.
- Structured the application to support future booking, itinerary management, and customer workflows through a scalable modular architecture.
- Applied full-stack development principles to establish reusable backend and frontend foundations for future feature expansion.

EDUCATION

K.S.R College of Engineering

Bachelor of Engineering (B.E.) in Computer Science and Engineering

Namakkal, Tamil Nadu, India

Aug 2019 – May 2023

CERTIFICATIONS

AWS DynamoDB Fundamentals – Meta | **Rust Programming Essentials** – Edureka |

Advanced React – Whizlabs | **Google Prompting Essentials** – Google